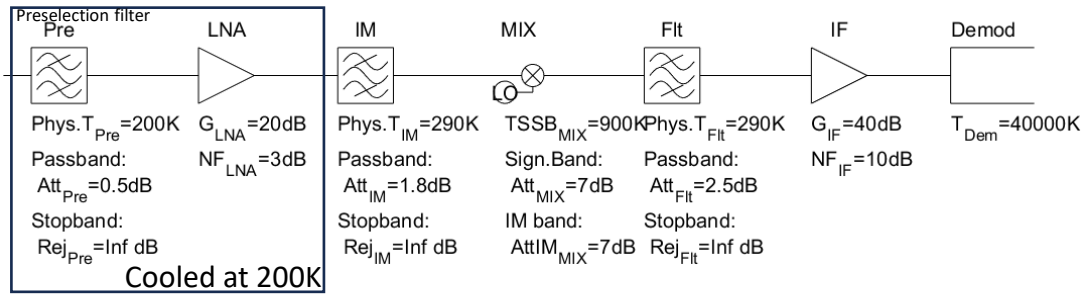
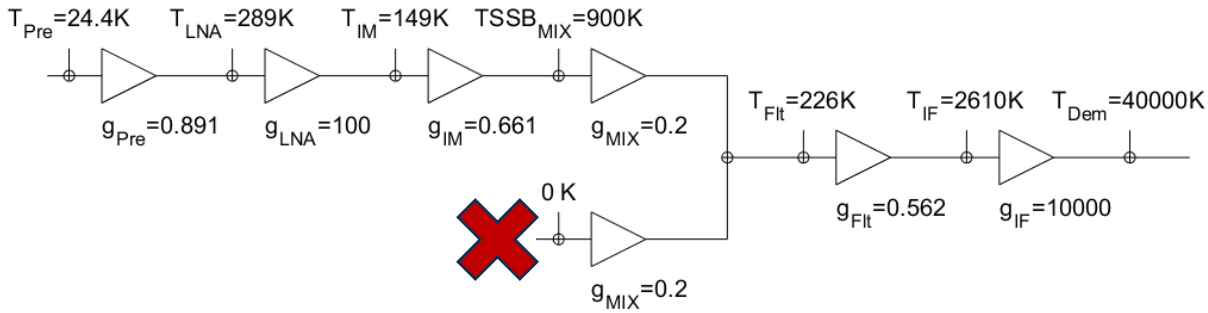
Exercise

Compute the Noise Figure of the overall receiver in the diagram below. Then indicate which is the component whose noise dominates the overall NF and a possible counteraction.



Solution

Draw the noise diagram and convert all gains and attenuations into gains in linear scale ($g=10^{GdB/10}$ or $g=10^{-AttdB/10}$) and all noise figures into equivalent noise temperatures:



Beware: the physical temperature is relevant only for resistive attenuators: $Teq=Tphys(1-10^{NFdB/10})$ whereas the NF is defined at 290K (even if the component is kept at the prescribed cooled temperature): $Teq=290K(1-10^{NFdB/10})$.

Then assemble all noise contributions in the equivalent noise temperature of the receiver:

$$\left(\frac{T_{eq}}{780K}\right) = \left(\frac{T_{Pre}}{24.4K}\right) + \frac{\left(\frac{T_{LNA}}{289K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\right)} + \frac{\left(\frac{T_{IM}}{149K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\cdot\left(\frac{g_{LNA}}{100}\right)\right)} + \frac{\left(\frac{TSSB_{MIX}}{900K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\cdot\left(\frac{g_{LNA}}{100}\right)\cdot\left(\frac{g_{IM}}{0.661}\right)\right)} + \frac{\left(\frac{T_{Flt}}{226K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\cdot\left(\frac{g_{LNA}}{100}\right)\cdot\left(\frac{g_{IM}}{0.661}\right)\cdot\left(\frac{g_{MIX}}{0.2}\right)\right)}$$

$$+ \frac{\left(\frac{T_{IF}}{2610K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\cdot\left(\frac{g_{LNA}}{100}\right)\cdot\left(\frac{g_{IM}}{0.661}\right)\cdot\left(\frac{g_{MIX}}{0.2}\right)\cdot\left(\frac{g_{Flt}}{0.562}\right)\right)} + \frac{\left(\frac{T_{Dem}}{40000K}\right)}{\left(\left(\frac{g_{Pre}}{0.891}\right)\cdot\left(\frac{g_{LNA}}{100}\right)\cdot\left(\frac{g_{IM}}{0.661}\right)\cdot\left(\frac{g_{MIX}}{0.2}\right)\cdot\left(\frac{g_{Flt}}{0.562}\right)\cdot\left(\frac{g_{IF}}{10000}\right)\right)}$$

The individual contributions are:

Teq=24.4K(Pre) + 324K(LNA)+1.7K(IM)+15.8K(MIX)+19K(Flt)+427K(IF)+0.6K(Dem)

The NF of the overall receiver is then:

$nf_{rec}=1+Teq/290K=3.7 \rightarrow NF_{rec}=5.7dB$

Based on the individual contributions, the highest comes from the IF amplifier.
One possible countermeasure would be to introduce another amplifier before (e.g., after the LNA) with a reasonably-low noise figure.